Theory of Elementary Particles
Lecture 13

Krzysztof M. Graczyk
kgraczyk.github.io

Institute of Theoretical Physics
University of Wrocław
pl. M. Borna 9, 50-204 Wrocław, Poland

05.01.2027, Wrocław



Cabibbo Angle

- Purely leptonic decay

$$\mu^- \to e^- \overline{\nu}_e \nu_\mu \tag{1}$$

as well as

$$e^- \nu_\mu \to \mu^- \nu_e \tag{2}$$

are described by Hamiltonian

$$H^l_{weak} = \frac{G_F}{\sqrt{2}} \left(\overline{\nu}_\mu \gamma^\alpha (1-\gamma_5)\mu\right) \left(\overline{e}\gamma_\alpha (1-\gamma_5)\nu_e\right) \tag{3}$$

- From the measurement of μ-life time

$$\tau_\mu = 2.197 \times 10^{-6}\,\mathrm{s} \tag{4}$$

one gets the value of decay coupling G_F

- β-decay is described within

$$H^h_{weak} = \frac{G^{(0)}}{\sqrt{2}} \left(\overline{p}\gamma^\alpha(1 - C_A\gamma_5)n\right)\left(\overline{e}\gamma_\alpha(1-\gamma_5)\nu_e\right) \tag{5}$$

- From experimental measurements:

$$\frac{G^0}{G_F} \approx 0.975 \tag{6}$$

$$C_A \approx 1.276 \tag{7}$$

- One would expect a similarity between the weak interactions of leptons and hadrons!
- Axial-Vector current coupling gets modified by the QCD effects causing C_A to deviate from 1!
- Electromagnetic current is conserved and also vector current is conserved so the matrix element of the vector part should not be renormalized at the hadron level.
- The difference between $G^{(0)}$ and G_F has a deeper physical meaning.

Remark

Weak ($\Delta S = 1$)

$$\Lambda \to p e \overline{\nu}_e \tag{8}$$

is described by Hamiltonian

$$H^{\Lambda}_{weak} = \frac{G^{(1)}}{\sqrt{2}} \left(\overline{p}\gamma^{\alpha}(1 - C_A\gamma_5)\Lambda\right)\left(\overline{e}\gamma_{\alpha}(1-\gamma_5)\nu_e\right) \tag{9}$$

but with

$$G^{(1)} = 0.22\, G_F \tag{10}$$

where

$$(0.975)^2 + (0.22)^2 \approx 1 \tag{11}$$

Hypothesis $G^{(1)} \sim \sin\theta_C G_F$ and $G^{(0)} \sim \cos\theta_C G_F$

$$\cos^2\theta_C + \sin^2\theta_C = 1 \tag{12}$$

- Notice that $p \sim uud$ and $\Lambda \sim uds$
- Hence, there is a transition $s \to u$ with the strength $\sin\theta_C G_F$

- Mixing between d and s quarks?
- On the quark level one may construct

$$J_\alpha^+ = \overline{u}\gamma_\alpha(1-\gamma_5)d_C, \tag{13}$$

where

$$d_C = d\cos\theta_C + s\sin\theta_C \tag{14}$$

θ_C Cabibbo angle.

Remark

Similar substitution $d \to d_C$ should work for neutral current interactions

$$\overline{d}\gamma_\alpha(g_V^d - \gamma_5 g_A^d)d \to \overline{d}_C\gamma_\alpha(g_V^d - \gamma_5 g_A^d)d_C \tag{15}$$

Then

$$\cos\theta_C \sin\theta_C[\overline{d}\gamma_\alpha(g_V^d - \gamma_5 g_A^d)s + \overline{s}\gamma_\alpha(g_V^d - \gamma_5 g_A^d)d] \tag{16}$$

- It suggests that the transition $s \to d$ $(\cos\theta_C \sin\theta_C)$ is comparable in strength to $s \to u$ $(\sin\theta_C)$.
- **But it is in sharp disagreement with the data, namely**:

$$s \to d \qquad K^- \to \pi^- e^+ e^- \tag{17}$$

should be comparable with observed

$$s \to u \qquad K^- \to \pi^0 e^- \overline{\nu} \tag{18}$$

- $\Delta S = 1$ Not allowed! $\overline{u}s \to \overline{u}d$, Z^0 emitted
- $\Delta S = 1$ Allowed!: $\overline{u}s \to \overline{u}u$, W^-
- NO **Flavor Changing Neutral Currents (FCNC)**

GIM mechanism

- It was suggested that there exists additional flavor of quark, charm c, which forms with s additional quark doublet, which is orthogonal to (u, d) doublet.
- The lower components of the two doublets

$$\begin{pmatrix} u \\ d_C \end{pmatrix}, \quad \begin{pmatrix} c \\ s_C \end{pmatrix} \tag{19}$$

are related to the physical quarks d and s by orthogonal transformation

$$\begin{pmatrix} d_C \\ s_C \end{pmatrix} = V_C \begin{pmatrix} d \\ s \end{pmatrix} = \begin{pmatrix} \cos\theta_C & \sin\theta_C \\ -\sin\theta_C & \cos\theta_C \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix} \tag{20}$$

where

$$V_C^T V_C = 1$$

Remark

As long as the coupling strength of two doublets to Z_0 are equal, the flavor changing neutral processes such as $s \to d$ are suppressed to all orders of perturbation theory because of orthogonality.

- Indeed from orthogonality we have:

$$\overline{d}_C d_C + \overline{s}_C s_C = \overline{d} d + \overline{s} s \tag{21}$$

  Hence no Cabibbo angle in neutral current interactions

- For charged current interaction there is

$$J_\mu(quarks) \quad = \quad \frac{1}{2}\overline{d}_C \gamma_\mu (1-\gamma_5) u + \frac{1}{2}\overline{s}_C \gamma_\mu (1-\gamma_5) c \tag{22}$$

  - Transition: $u \to d$ with $\cos\theta_C$
  - Transition: $u \to s$ with $\sin\theta_C$
  - Transition: $c \to d$ with $\sin\theta_C$
  - Transition: $c \to s$ with $\cos\theta_C$

$$
\begin{aligned}
J_0^\mu &= \frac{1}{2}\overline{u}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)u + \frac{1}{2}\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d && (23)\\
&\to \frac{1}{2}\overline{u}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)u + \frac{1}{2}\overline{d}_C\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d_C && (24)\\
&+ \frac{1}{2}\overline{c}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)c + \frac{1}{2}\overline{s}_C\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s_C && (25)\\
&= \frac{1}{2}\overline{u}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)u + \frac{1}{2}\overline{c}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)c && (26)\\
&\quad + \frac{1}{2}\cos^2\theta_C\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d + \frac{1}{2}\sin^2\theta_C\overline{s}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s && (27)\\
&\quad + \frac{1}{2}\cos\theta_C\sin\theta_C\left[\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s + \overline{s}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d\right] && (28)\\
&\quad + \frac{1}{2}\sin^2\theta_C\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d + \frac{1}{2}\cos^2\theta_C\overline{s}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s && (29)\\
&\quad - \frac{1}{2}\cos\theta_C\sin\theta_C\left[\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s + \overline{s}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d\right] && (30)\\
&= \frac{1}{2}\overline{u}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)u + \frac{1}{2}\overline{c}\gamma^\mu\left(g_V^u - g_A^u\gamma_5\right)c && (31)\\
&\quad + \frac{1}{2}\overline{d}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)d + \frac{1}{2}\overline{s}\gamma^\mu\left(g_V^d - g_A^d\gamma_5\right)s && (32)
\end{aligned}
$$

flavor mixing channels are not allowed!

Mass matrix

- The presence of the Cabibbo mixing indicates that one must distinguish between two types of fermionic states:
  - gauge interaction eigenstates (having definite gauge transformation properties, e.g. d_C, s_C).
  - mass eigenstates e.g. d and s fields

  they are related by some linear transformation.

Let us extend the lepton and quark families

- $$\begin{aligned} e &\rightarrow e'_A = (e',\, \mu',\, \tau') && (33)\\ \nu_e &\rightarrow \nu'_A = (\nu'_e, \nu'_\mu, \nu'_\tau) && (34)\\ u &\rightarrow p'_A = (u', c', t') && (35)\\ d &\rightarrow n'_A = (d', s', b') && (36) \end{aligned}$$

- $$l_{AL} = \begin{pmatrix} \nu'_A \\ e'_A \end{pmatrix}_L, \quad q_{AL} = \begin{pmatrix} p'_A \\ n'_A \end{pmatrix}_L \qquad (37)$$

- the prime on the field indicates it is gauge eigenstate.

The generalization of the Lagrangian (before symmetry breaking)

$$\mathcal{L}_2 = \overline{l_{AL}}\, i\gamma^\mu \left(\partial_\mu - \frac{ig}{2}\boldsymbol{\tau}\cdot\boldsymbol{A}_\mu + \frac{ig'}{2}B_\mu\right) l_{AL} + \overline{e'_{AR}}\, i\gamma^\mu \left(\partial_\mu + ig' B_\mu\right) e'_{AR} \quad (38)$$

$$+\overline{q_{AL}}\, i\gamma^\mu \left(\partial_\mu - \frac{ig}{2}\boldsymbol{\tau}\cdot\boldsymbol{A}_\mu - \frac{ig'}{6}B_\mu\right) q_{AL} \quad (39)$$

$$+\overline{p'_{AR}}\, i\gamma^\mu \left(\partial_\mu - \frac{2ig'}{3}B_\mu\right) p'_{AR} + \overline{n'_{AR}}\, i\gamma^\mu \left(\partial_\mu + \frac{ig'}{3}B_\mu\right) n'_{AR} \quad (40)$$

$$\mathcal{L}_4 = -f^{(e)}_{AB}\overline{l_{AL}}\Phi e'_{BR} - f^{(p)}_{AB}\overline{q_{AL}}\tilde{\Phi} p'_{BR} - f^{(n)}_{AB}\overline{q_{AL}}\Phi n'_{BR} + h.c. \quad (41)$$

- Notice that $f^{(e)}_{AB}$, $f^{(n)}_{AB}$, $f^{(p)}_{AB}$ are the matrices!
- Members of new families have the same quantum numbers as electron ν_e and u, d fermions!

After symmetry breaking

$$\begin{aligned}\mathcal{L}_4 &= -\frac{\eta}{\sqrt{2}}\left(f^{(e)}_{AB}\overline{e'_{AL}}e'_{BR}+f^{(n)}_{AB}\overline{n'_{AL}}n'_{BR}+f^{(p)}_{AB}\overline{p'_{AL}}p'_{BR}\right)\\ &\quad -\frac{v}{\sqrt{2}}\left(f^{(e)}_{AB}\overline{e'_{AL}}e'_{BR}+f^{(n)}_{AB}\overline{n'_{AL}}n'_{BR}+f^{(p)}_{AB}\overline{p'_{AL}}p'_{BR}\right)+h.c.\end{aligned}$$

in the gauge-eigenstate basis the fermionic mass matrices are

$$M^{(i)}_{AB}=\frac{v}{\sqrt{2}}f^{(i)}_{AB},\quad i=e,p,n \tag{42}$$

Remark

To have mass eigenstates we need to diagonalize $M^{(i)}_{AB}$. How to make it?

Biunitary transformation

Remark

There is no reason for the mass matrices to be diagonal. In general they are neither symmetric nor hermitian.

Theorem

For the mass matrix M there exists unitary matrices S and T such that

$$S^\dagger M T = M_d, \tag{43}$$

where M_d is diagonal matrix with positive eigenvalues.

In principle

- any matrix M can be always written as product of hermitian matrix H and unitary matrix V.
  - H-hermitian matrix can be diagonalized by some unitary matrix

Proof

- $M^\dagger M$ is hermitian, so it can be diagonalized
- Exists S so that

$$S^\dagger M M^\dagger S = M_d^2 \tag{44}$$

* $MM^\dagger$ is positive definite (for invertible M) i.e. for any $v \neq 0$, $v^\dagger M M^\dagger v > 0$, hence it has positive eigenvalues

- Certainly S matrix is unique up to diagonal phase matrix
-

$$(SF)^\dagger M M^\dagger (SF) = M_d^2 \tag{45}$$

with

$$F = \begin{pmatrix} e^{i\phi_1} & & \\ & e^{i\phi_2} & \\ & & e^{i\phi_3} \end{pmatrix} \tag{46}$$

- with the help of these phases we may fix the elements of M_d to be positive, namely

$$\begin{pmatrix} m_1 & & \\ & m_2 & \\ & & m_3 \end{pmatrix}, \quad m_i > 0 \tag{47}$$

- Define Hermitian matrix H by

$$H = SM_dS^\dagger \tag{48}$$

- Define the matrix

$$V \equiv H^{-1}M, \quad V^\dagger = M^\dagger H^{-1} \tag{49}$$

which is unitary, indeed,

$$\begin{aligned} VV^\dagger &\equiv H^{-1}M(H^{-1}M)^\dagger = H^{-1}MM^\dagger H^{-1} && (50)\\ &= H^{-1}SM_d^2S^\dagger H^{-1} && (51)\\ &= H^{-1}SM_dS^\dagger SM_dS^\dagger H^{-1} && (52)\\ &= H^{-1}HHH^{-1} = I && (53) \end{aligned}$$

- From the definition of H and V we have

$$S^\dagger H S = S^\dagger M V^\dagger S = M_d \tag{54}$$

or

$$S^\dagger M T = M_d \tag{55}$$

where $T = V^\dagger S$ is also unitary. Indeed,

$$V^\dagger S (V^\dagger S)^\dagger = V^\dagger S S^\dagger V = 1 \tag{56}$$

Biunitary transformation

- Consider the mass term

$$\overline{\psi'_L} M \psi'_R = (\overline{\psi'_L} S) S^\dagger M T (T^\dagger \psi'_R) = \overline{\psi_L} M_d \psi_R \tag{57}$$

with

$$\psi'_L = S\psi_L \tag{58}$$
$$\psi'_R = T\psi_R \tag{59}$$

- **Notice that the Left-handed and Right-handed fields transform in the different way...**

Charged current interactions

$$J_{q\mu} = \overline{q_{AL}}\gamma_\mu \tau^+ q_{AL} = \overline{p'_{AL}}\gamma_\mu n'_{AL} = \overline{p_{AL}}\gamma_\mu \left(S^\dagger_{(p)} S_{(n)}\right) n_{AL} \tag{60}$$

where

$$p'_L = S_{(p)} p_L \tag{61}$$
$$n'_L = S_{(n)} n_L \tag{62}$$

Thus in terms of mass eigenstates, the quark doublets of three families are

$$q_{AL}: \quad \begin{pmatrix} u \\ d'' \end{pmatrix}_L \quad \begin{pmatrix} c \\ s'' \end{pmatrix}_L \quad \begin{pmatrix} t \\ b'' \end{pmatrix}_L \tag{63}$$

where

$$\begin{pmatrix} d'' \\ s'' \\ b'' \end{pmatrix} = U \begin{pmatrix} d \\ s \\ b \end{pmatrix} \tag{64}$$

and

$$U = S^\dagger_{(p)} S_{(n)} \tag{65}$$

It is a mixing matrix! In the quark sector!

Neutral Current

$$
\begin{aligned}
J^{NC}_{q\mu} &\sim \overline{q_{AL}}\gamma_\mu(...)q_{AL} && (66)\\
&= \overline{p'_{AL}}\gamma_\mu(...)p'_{AL} + \overline{n'_{AL}}\gamma_\mu(...)n'_{AL} && (67)\\
&= \overline{p_{AL}}\gamma_\mu\left(S^\dagger_{(p)}S_{(p)}\right)p_{AL} + \overline{n_{AL}}\gamma_\mu\left(S^\dagger_{(n)}S_{(n)}\right)n_{AL} && (68)\\
&= \overline{q_{AL}}\gamma_\mu(...)q_{AL} && (69)
\end{aligned}
$$

- No mixing matrix in NC sector.

CP violation phase

- If the mixing matrix U is real, then U is an orthogonal matrix, hence all fermion gauge couplings are real and they do not induce CP violation (show in exercise)

$$(\mathcal{CP})J_\mu^+(\mathcal{CP})^{-1} = (\mathcal{CP})\overline{p_{AL}}\gamma^\mu U n_{AL}(\mathcal{CP})^{-1} \quad = \quad -\overline{n_{AL}}U^T\gamma^\mu p_{AL} \tag{70}$$

  Notice that

$$\overline{\Psi_L}\gamma_\mu\psi_L = \frac{1}{4}\overline{\Psi}\gamma_\mu(1-\gamma_5)^2\psi = \frac{1}{2}\overline{\Psi}\gamma_\mu(1-\gamma_5)\psi \tag{71}$$

  and

$$(\overline{\Psi}\gamma_\mu(1-\gamma_5)\psi)^\dagger = \psi^\dagger(1-\gamma_5)\gamma_\mu^\dagger\gamma_0\Psi = \overline{\psi}\gamma_\mu(1-\gamma_5)\Psi \tag{72}$$

- Arbitrary unitary matrix U can be parametrized by the **three** real rotational angles and **six** phases (in total 9 real parameters)
- Some of the phases have physical meaning but some of them can be removed...

- five phases can be removed by redefinition of quark fields

$$q_{1L} = \begin{pmatrix} u \\ U_{11}d + U_{12}s + U_{13}b \end{pmatrix} \tag{73}$$

let $U_{11} = R_{11}e^{i\delta}$ where R_{11} is real then δ phase can be eaten by redefinition $u' \to ue^{-i\delta}$ hence

$$q_{1L} = e^{i\delta} \begin{pmatrix} u' \\ R_{11}d + U'_{12}s + U'_{13}b \end{pmatrix} \tag{74}$$

- similarly we factor out the complex phases of U'_{21} and U'_{31} by redefinition of c and t quark fields
- eventually we factor out the phases of fields s and b

$$\begin{pmatrix} u' \\ R_{11}d + R_{12}s' + R_{13}b' \end{pmatrix}_L \quad \begin{pmatrix} c' \\ R_{21}d + R_{22}e^{i\delta_1}s' + R_{23}e^{i\delta_2}b' \end{pmatrix}_L \tag{75}$$

$$\begin{pmatrix} t' \\ R_{31}d + R_{32}e^{i\delta_3}s' + R_{33}e^{i\delta_4}b' \end{pmatrix}_L \tag{76}$$

- Now we have four phases and 9 R_{ij} real parameters

- we left with 9 R_{ij} parameters and 4 phases δ_i
- normalization of each state gives three conditions

$$\overline{d}_C d_C \sim 1 \quad (77)$$
$$\overline{s}_C s_C \sim 1 \quad (78)$$
$$\overline{b}_C b_C \sim 1 \quad (79)$$

- orthogonality conditions among the different states yield six conditions

$$\overline{d}_C s_C = 0,\ \overline{d}_C b_C = 0,\ \overline{b}_C s_C = 0$$

  each of the three conditions is complex, hence 6 real constraints
- After reduction there is one phase and 3 real parameters!
- Remaining mixing matrix U is called Cabibbo, Kobayashi and Maskawa (CKM) matrix!

Two families: no CP violation!

- Two quark families

$$\begin{pmatrix} u \\ U_{11}d + U_{12}s \end{pmatrix}, \quad \begin{pmatrix} c \\ U_{21}d + U_{22}s \end{pmatrix} \Rightarrow \begin{pmatrix} u \\ R_{11}d + U_{12}s \end{pmatrix}, \quad \begin{pmatrix} c \\ R_{21}d + U_{22}s \end{pmatrix}$$

Now we redefine s then

$$\begin{pmatrix} u \\ R_{11}d + R_{12}s \end{pmatrix}, \quad \begin{pmatrix} c \\ R_{21}d + U_{22}s \end{pmatrix}$$

- From normalization and orthogonality we have

$$(R_{11})^2 + (R_{12})^2 = 1 \tag{80}$$
$$(R_{21})^2 + (R_{22})^2 = 1 \tag{81}$$

and

$$R_{11}R_{21} + R_{12}R_{22}e^{i\phi} = 0 \Rightarrow \phi = 0 \tag{82}$$

- there is only one angle and **no phase** hence in two family scenario there is no CP violation

- CKM matrix

$$U = \begin{pmatrix} U_{ud} & U_{us} & U_{ub} \\ U_{cd} & U_{cs} & U_{cb} \\ U_{td} & U_{ts} & U_{tb} \end{pmatrix} \tag{83}$$

- Or written in terms of angles

$$U = \begin{pmatrix} c_1 & s_1 c_3 & s_1 s_3 \\ -s_1 c_2 & c_1 c_2 c_3 - s_2 s_3 e^{i\delta} & c_1 c_2 s_3 + s_2 c_3 e^{i\delta} \\ -s_1 s_2 & c_1 s_2 c_3 + c_2 s_3 e^{i\delta} & c_1 s_2 s_3 - c_2 c_3 e^{i\delta} \end{pmatrix} \tag{84}$$

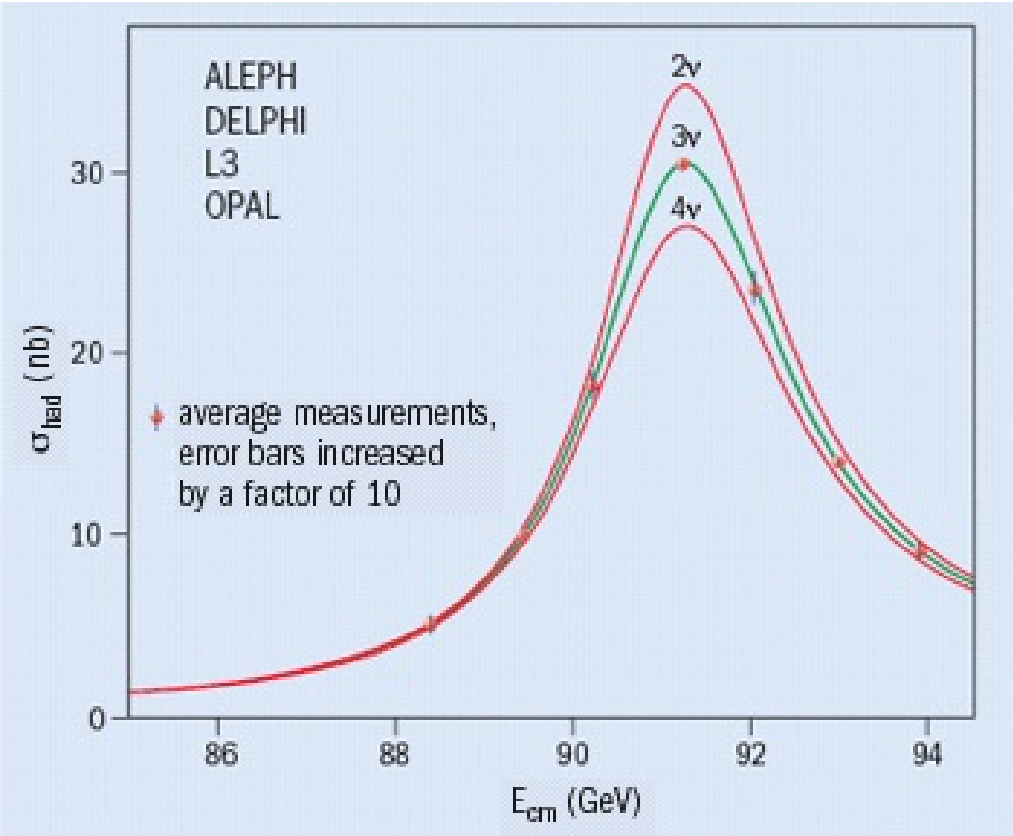
  where $0 \leq \theta_i \leq \pi/2$, $-\pi \leq \delta < \pi$

- $s_i \equiv \sin\theta_i$, $c_i \equiv \cos\theta_i$
- θ_1 corresponds closely to the Cabibbo angle; Cabibbo universality implies $\theta_3 \approx 0$
- From β decay

$$|U_{ud}| = 0.97367 \pm 0.00032 \tag{85}$$

- From Hyperon decay

$$|U_{us}| = 0.2243 \pm 0.0008 \tag{86}$$



- Number of light neutrino species from the Z^0 line shape (LEP):

$$N_\nu = 2.9840 \pm 0.0082$$